QQ-A-671A
March 3, 1972
SUPERSEDING
Fed. Spec. QQ-A-671
September 3, 1957

FEDERAL SPECIFICATION

ANODES, CADMIUM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

* 1.1 Scope. This specification covers the requirements for one grade of cadmium anodes used in electroplating baths.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

PPP-B-585	-	Boxes; Wood, Wire-Bound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)
Fed. Test Method Std.
No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

FSC 3426

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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Materials. Anodes shall be of solid cadmium. They shall be either cast, rolled to size, or cut to size from electrorefined metal, castings or rolled plate.

* 3.2 Size,shape and method of suspension. The size, shape and method of suspension (when applicable) of the anode shall be as specified (see 6.2). Ball anodes shall have a diameter of approximately two inches and weight not less than 1-1/4 pounds per ball. Bar anodes, of elliptical or oval cross section, when furnished in lengths of 18 to 96 inches, shall weigh not less than 1-1/2 pounds per linear inch. Sheared bars and shapes with one or both ends hemispherical are acceptable provided the overall length does not exceed four inches with a maximum diameter of two inches.

* 3.3 Chemical composition. Unless otherwise specified in the contract or order (see 6.2), the cadmium anodes shall meet the chemical requirements shown in Table I.

* 3.4 Identification. Each anode, except ball anodes, shall be permanently identified with the following information:

(a) "Anode, Cadmium"
(b) Specification Number

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TABLE I

CHEMICAL COMPOSITION

Element	Percent Composition
Cadmium 1/	99.90, Min.
Zinc	0.035, Max.
Lead	0.025, Max.
Copper	0.015, Max.
Silver	0.01, Max.
Tin	0.01, Max.
Arsenic	0.003, Max.
Thallium	0.003, Max.
Antimony	0.001, Max.

1/ Cadmium shall be determined by difference

3.4.1 Ball anodes. Each ball anode shall be permanently marked with the word "Cadmium" or the symbol "Cd", either as the result of casting or otherwise imprinting the selected identifying mark.

3.5 Workmanship. Anodes shall be clean and substantially free from cracks, warps, inclusions, porosity, ragged edges, surface films such as rolling skin, and other defects which may adversely affect uniform corrosion in service. Anodes shall not be broken.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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* 4.2 Lot. Unless otherwise specified in the contract or order (see 6.2), a lot shall consist of all anodes of one shape in a single shipment. When practicable, each lot shall be made up of anodes of the same melt, size and weight.

* 4.3 Quality conformance sampling.

* 4.3.1 For visual examination and nondestructive tests. Unless otherwise specified (see 6.2), samples for visual examination and nondestructive tests shall be selected at random from each lot in accordance with MIL-STD-105, Level II, Acceptable Quality Level (AQL) 1.5 percent defective.

* 4.3.2 For chemical analysis. Sample anodes shall be selected as follows:

(a) For ball anodes or sheared bars and shapes not exceeding four inches in length, at least 4 sample anodes shall be selected from each lot or from each melt if the lot consists of more than one melt. If the melt or melts cannot be identified, at least 8 anodes shall be selected at random from each lot.

(b) For shapes other than balls or sheared bars and shapes not exceeding four inches in length, at least 2 sample anodes shall be selected at random from each lot or from each melt, if the lot consists of more than one melt. If the melt or melts cannot be identified, at least 4 anodes shall be selected at random from each lot.

* 4.3.2.1 Preparation for chemical analysis. A composite sample of at least 6 ounces of chips or drillings shall be obtained by drilling, sawing, milling or machining the sample anodes of 4.3.2 in accordance with Fed. Test Method Std. No. 151, Method 111. No lubricant shall be used in the operation. The sample shall be free from dirt, oil,grit and other foreign matter. Chips or drillings shall be treated with a magnet to remove any steel introduced in the preparation of the sample. Anodes which have been drilled or machined for chemical analysis may be included in the lot for shipment.

* 4.3.2.2 Preparation for spectrochemical analysis. A composite sample shall be prepared as detailed in 4.3.2.1 or rod electrodes shall be prepared from each of the sample anodes of 4.3.2 in accordance with Fed. Test Method Std. No. 151, Method 112. The rod samples shall be prepared by conventional methods, suitable for the correct applications and usages of the techniques employed. No lubricant shall be used in the preparation operation. The samples shall be free from dirt, oil, grit and other foreign matter. Anodes, except for the ball type, from which electrodes have been obtained may be included in the lot for shipment.

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* 4.4 Quality conformance examination.

* 4.4.1 Anodes. Sample anodes, selected in accordance with 4.3.1, shall be visually inspected, measured and weighed for conformance to the requirements of 3.2, 3.4 and 3.5. If the number of defective items in any sampling exceeds the acceptance number for the specified sample size, the lot represented by the sample shall be rejected.

* 4.4.2 Packing and marking. The packing and marking of anodes furnished under this specification shall be inspected for conformance to the applicable requirements of Section 5.

* 4.5 Quality conformance testing.

* 4.5.1 Chemical composition. Sample anodes, selected in accordance with 4.3.2, shall be tested in accordance with Method 111 or Method 112 of Fed. Test Method Std. No. 151 to determine conformance to the chemical requirements of Table I. If the sample fails to conform to the requirements of Table I, the lot shall be rejected. Standard methods of analysis shall be used (see 6.3).

5. PREPARATION FOR DELIVERY

* 5.1 Packing. When practicable, anodes of the same size and shape shall be packed together. Unless otherwise specified, the gross weight of each box shall not exceed approximately 200 pounds (see 6.2).

* 5.1.1 Level A. Anodes shall be packed in overseas type boxes in accordance with PPP-B-601 or PPP-B-621.

5.1.2 Level B. Anodes shall be packed in domestic type boxes in accordance with PPP-B-585, PPP-B-601 or PPP-B-621.

* 5.1.3 Level C. Anodes shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Containers shall meet, as a minimum, the requirements of rules and regulations applicable to the mode of transportation selected.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.2), shipping containers shall be marked in accordance with FED-STD-123.

* 5.2.2 Military agencies. In addition to any special markings required by the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

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* 5.2.3 Special markings. Each container shall be marked with the following:

(a) Specification Number
(b) Nomenclature of the anode, as applicable
(c) Name of contractor
(d) Contract Number or Order Number
(e) Quantity
(f) Gross weight

6. NOTES

* 6.1 Intended use. Cadmium anodes are intended for use in cadmium-plating baths. Ball-shaped anodes, in a spiral cage of bare steel, are satisfactory for plating cadmium from a cyanide bath. However, ball anodes, in uncoated steel cages should not be used when cadmium is plated from an acid bath, such as the fluoborate. Either bar anodes or bagged cast anodes should be employed. Bar anodes should not be used in cyanide baths because they must be removed and replaced when the cross sectional area decreases or will dissolve at the solution level and drop to the bottom of the tank.

* 6.2 Ordering data. Purchasers should select the perferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.

(b) Size, shape and method of suspensions (when applicable) (see 3.2).

(c) Chemical composition, if different from Table I (see 3.3).

(d) Size of lot, if different from 4.2.

(e) Sampling procedures if different from 4.3.1.

(f) Gross weight of packing, if different from 5.1.

(g) Applicable level of packing (see 5.1.1, 5.1.2 and 5.1.3).

(h) Additional marking, if necessary (see 5.2).

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* 6.3 Method of chemical analysis. It is intended that the chemical analysis be conducted so that no particular method of analysis be specified. Metallic impurities in cadmium anodes may be determined by spectrographic methods or photometric methods. As a guide, ASTM E396, Standard Method for Chemical Analysis of Cadmium Metal may be used for determination of the lead, copper, arsenic and thallium contents by photometric methods.

Military Custodians
Army - WC
Navy - AS
Air Force - 84

Review activities
Army - None
Navy - None
Air Force - None
Defense Supply Agency - GS

User activities
Army - GL, AT
Navy - SH, OS
Air Force - None

Civil Agencies Interest
GSA - FSS

Preparing activity
Navy - AS
(Proj. No. 3426-0009)

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